



IDBI INNOVATE 2026

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Team Details

- a. Team name: Vizagites
- b. Team leader name: T.A.Srinivas
- c. Problem Statement: Financial Health Score

Track 3 - Financial Health Score Repository : MSME Financial Health Passport

What is the MSME Credit Health Portal?

- **The Big Picture:** Small businesses frequently struggle to get loans because they lack traditional credit histories. This portal bridges that gap by assessing an MSME's real-time financial health using their day-to-day digital footprint.
- **No Black Boxes:** Instead of spitting out a random, confusing risk number, the portal relies on Explainable AI (XAI). It explicitly tells the lender and the business owner exactly what behaviors are boosting or hurting their score.
- **Democratic Underwriting:** The system evaluates three modern alternative data layers: cash flow liquidity from banks, GST tax records, and daily digital operations like UPI transactional history and EPFO employee salary punctuality.

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Bridging the Credit Gap

- **The Untapped Market:** Millions of New-To-Credit (NTC) and New-To-Banking (NTB) micro-enterprises are completely invisible to standard lending models.
- **Alternative Data Boom:** The framework easily harnesses modern digital infrastructure—such as open banking networks, the GSTN portal, and real-time payment protocol interfaces.
- **Frictionless Approvals:** It cuts down loan processing times from several weeks of manual document checking to a matter of single-click digital assessments.
- **Dynamic Credit Monitoring:** Lenders move away from stale, annual point-in-time reviews to real-time risk updates as the merchant's business shifts.

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Core Features Built into the Platform

- **Omni-Reactive Data Panel:** Instantly syncs raw uploaded data records directly with backend scoring mechanics.
- **Interactive What-If Sliders:** Allows users to simulate how improving a metric (like reducing tax delay days) directly increases their final credit rating.
- **XGBoost Risk Grading:** High-speed algorithmic scoring that predicts default probability based on verified data contrast features.
- **SHAP Visualizations:** Displays clear horizontal bar charts highlighting exact positive and negative financial performance drivers.
- **Dynamic Portfolio Splitters:** Segment customers instantly into "Approved Portfolio" or "Rejected Portfolio" downloadable subsets for batch processing.
- **Audit-Ready Passports:** Generates automated, secure PDF summaries stamped with digital protocol aggregator verifications.

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1. Data Ingestion

- Live Aggregator Inflow
- Sandbox CSV Upload

2. Analytics Engine

- XGBoost Risk Predictive
- SHAP Driver Breakdown

3. Output Distribution

- Dynamic Underwriting Tiers
- Instant PDF Passports

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The Proposed
Solution: Automated
Real-Time Risk
Assessment Interface



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AI-Driven MSME Financial Health Passport

Designed for TRACK 03: Financial Health Score=Financial Inclusion, Digital Lending, Credit Decisioning

- This dashboard translates alternate business metrics (GST, UPI, Bank Records) into an instant credit decision tool.
- Built as an accessible passport framework that anyone can easily interpret.

Data Ingestion Protocol Selection

Select Ingest Ingestion Channel:

- ☒ Live Ecosystem APIs (BIL / GSTN / AA Simulation)
- ☐ Batch Document Upload (CSV Sandbox)

Step 1: Select Active Row & Fine-Tune Parameters

Choose target MSME to inspect:

MSME-0001

Assign Corporate Display Name

Account Aggregator Records

Average Daily Balance kept in Bank (INR)

0

Money Inflow vs Outflow Ratio

0.00

Cheque Bounces due to low funds (INR)

0

Tax & Sales Records (GST Portal)

Average Monthly Sales/Turnover (INR)

0

Dependency Risk (High = Single Buyer)

0.00

Average Tax Filing Delay (Days)

0

Everyday Digital Operations (UPI & EFTPS)

Total UPI Sales Transactions per Month

0

Average Bill Amount per Customer (INR)

10

Active Registered Staff Count

1

Staff Fund Payment Timeliness (1.0 = Perfect)

0.00

Active Registry Status

Running on baseline Synthetic Databank Engine (1,200 Rows). Connected via ULI architectures.

Step 2: Live Credit Card Passport Results

SYSTEM ASSESSMENT: **EXCELLENT FINANCIAL HEALTH**

Your Financial Health Score: **855 / 900**

Estimated Risk Level: **7.41%**

Plain English Attributions (Explainable AI)

Cheque Bounces (Low Funds)

Staff Salary Fund Punctuality

Risk of Depending on Single Buyer

Average Customer Bill Size

Average Bank Balance

Employees Headcount Size

Monthly Sales Turnover

Monthly UPI Sales Volume

Money In vs Money Out Ratio

Tax Filing Delay Days

Risk Contribution Weight

Master Data Export Controls

Download Approved Portfolio (1020 Rows)

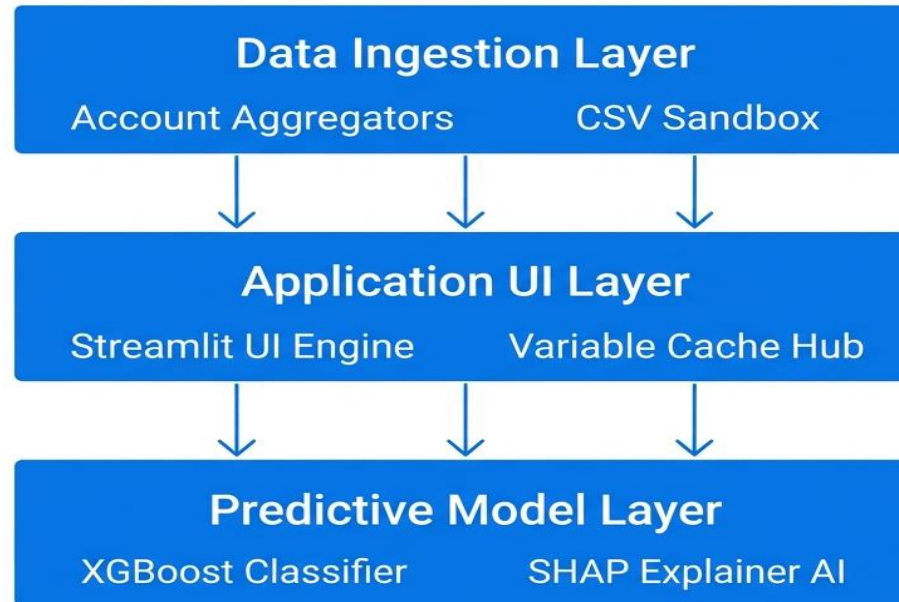
Download Rejected Portfolio (280 Rows)

Export Specific Client Document

TRACK 03 EXPECTED OUTCOME: Generates an Instant Financial Health Card tailored specifically for individual RTG/NTB applicants.

Download Customized PDF Passport for Sri Venkateswara Enterprises (MSME0001)

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Modern Python-Centric Tech Stack

- **Frontend Web App UI:** Streamlit — Enables high-velocity prototype layout building and real-time interactive dashboards.
- **Data Processing & Math:** Pandas & NumPy — Handles missing value fillna fallbacks, matrix conversions, and statistical percentiles.
- **Machine Learning Engine:** XGBoost — Built with domain risk monotone constraints natively enforced to handle strict regulatory credit alignment.
- **Explainable AI Framework:** SHAP — Extracts exact data impacts to break down the predictive model logic for non-technical users.
- **Enterprise Reporting Engine:** ReportLab — Assembles data tables, custom Hex typography styles, and grid layouts directly into secure, un-editable output files.

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Financial and Asset Breakdown

- **Development Phase Cost:** The open-source code foundation significantly lowers starting costs. Operational expenses are driven purely by low-tier application cloud server instances (e.g., Streamlit Community Cloud or basic AWS EC2).
- **API Integration Allocation:** Future budgets will shift primarily to licensing access keys for live verification networks (e.g., GSTN endpoints and Account Aggregator gateway nodes).
- **Prototype Sanity Stamping:** The running alpha uses built-in data generators that spin up balanced 1,200-row synthetic databanks instantly if a custom sheet is missing, allowing offline testing anywhere.

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Technical Benchmarking Metrics

- **Extreme Scoring Speed:** Predictions and feature impact outputs render in **less than 250 milliseconds** per unique merchant file, enabling true real-time calculations.
- **Native Guardrail Protections:** Built-in string-to-numeric coercion layers automatically clean up poorly formatted user spreadsheets, converting messy symbols to safe floats to prevent app crashes.
- **Robust Domain Alignment:** Monotone split constraints prevent systemic machine learning anomalies. For instance, a rise in cheque bounces is mathematically forced to lower the credit rating.
- **Data Contrast Fallback:** Automatic single-row edge-case modifiers guarantee that the engine remains stable even when a user uploads incredibly sparse or tiny customer lists.

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Moving from Prototype to Production

- **Core Ledger Integration:** Hook the portal natively into live banking core systems (like Finacle or cloud banking APIs) for direct corporate loan origination.
- **On-Device Local AI:** Transition background model configurations into compressed local setups so regional loan officers can run assessments in remote offline locations.
- **Real-time Fraud Triggers:** Add anomalies checks to flag businesses that artificially inflate their monthly UPI counts right before applying for a loan.

Proposed Project Links:

- [GitHub Public Repository](#)
- [Demo Video Link](#)
- [Final Product Link](#)



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Thank you!